

DATA DICTIONARY

MERCHANDISE PLANNING & ASSORTMENT OPTIMIZATION

MODEL SCOPE

2,255 SKUs - 10 stores - 120,000 sales records

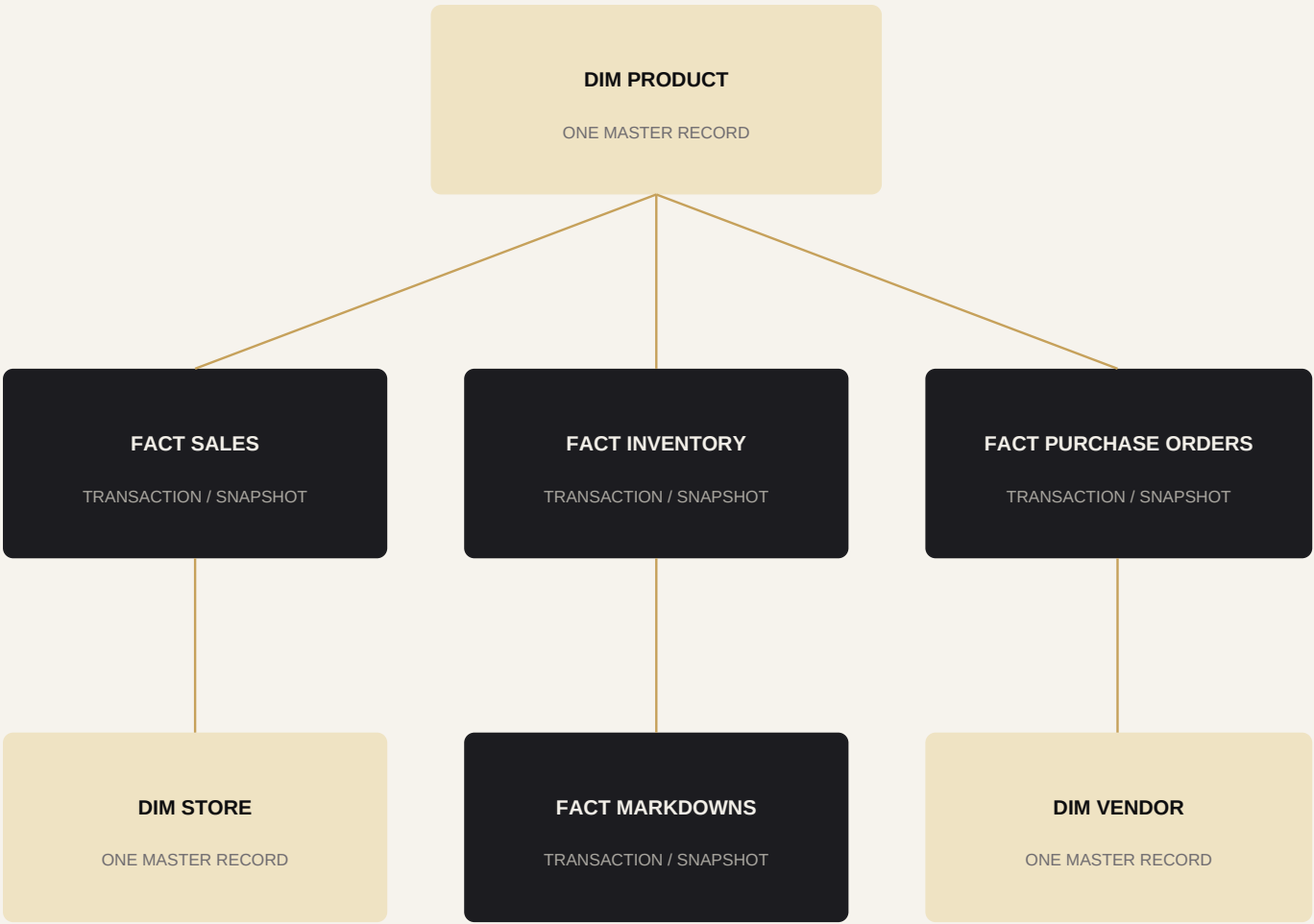
Seven-table retail merchandising star schema

Synthetic portfolio dataset - as of 12 August 2026

Data Model Overview

Business grain, relationships and modelling conventions

STAR SCHEMA



CONVENTIONS

Primary keys use stable text identifiers. Monetary values are stored in Canadian dollars. Rates are decimals between zero and one. Dates use ISO YYYY-MM-DD format. Sales is transaction-line grain; inventory is a dated store-SKU snapshot.

dim_product

Field definitions and business use

FIELDS

FIELD	TYPE	DEFINITION
SKU_ID	Text	Unique stock keeping unit identifier
Product_Name	Text	Merchandise description
Department / Category / Subcategory	Text	Retail product hierarchy
Vendor_ID	Text	Supplier key
Product_Cost	Currency	Unit landed product cost
Original_Retail_Price	Currency	Initial ticket retail
Current_Retail_Price	Currency	Active selling price
Margin_Percentage	Percent	Initial retail margin rate
Launch_Date	Date	First assortment date
Season	Text	Core or seasonal group
Product_Status	Text	Active, Seasonal or Exit
Price_Band	Text	Entry, Good, Better or Best

fact_sales

Field definitions and business use

FIELDS

FIELD	TYPE	DEFINITION
Transaction_ID	Text	Unique transaction-line key
Date	Date	Sales transaction date
Store_ID / SKU_ID	Text	Store and product foreign keys
Units_Sold	Integer	Quantity sold
Gross_Sales	Currency	Original retail value
Discount	Currency	Promotional and markdown reduction
Net_Sales	Currency	Gross sales less discount
Product_Cost	Currency	Cost of goods sold
Gross_Margin	Currency	Net sales less product cost
Markdown_Flag	Boolean	Markdown transaction indicator

fact_inventory

Field definitions and business use

FIELDS

FIELD	TYPE	DEFINITION
Date	Date	Snapshot date
Store_ID / SKU_ID	Text	Store-product grain
Beginning_Inventory	Integer	Opening inventory units
Units_Received	Integer	Receipts during period
Units_Sold	Integer	Units sold during period
Units_Transferred_In / Out	Integer	Inter-store inventory movements
Damaged_Units	Integer	Unsellable units
Ending_Inventory	Integer	Closing inventory units
Inventory_Value	Currency	Ending units at product cost

dim_store / dim_vendor

Field definitions and business use

FIELDS

FIELD	TYPE	DEFINITION
Store_ID	Text	Unique store key
City / Province / Region	Text	Store geography
Store_Format / Sales_Tier	Text	Operating segment
Inventory_Capacity	Currency	Planned inventory capacity
Vendor_ID	Text	Unique supplier key
Average_Lead_Time	Integer	Average supply lead time in days
On_Time_Delivery_Percentage	Percent	On-time delivery performance
Defect_Rate	Percent	Share of units received defective

fact_purchase_orders / fact_markdowns

Field definitions and business use

FIELDS

FIELD	TYPE	DEFINITION
PO_Number	Text	Unique purchase order line
Expected / Actual Delivery Date	Date	Promised and received dates
Units_Ordered / Received	Integer	Purchase order quantities
Delivery_Status	Text	On Time or Late
Markdown_ID	Text	Unique markdown event
Markdown_Percentage	Percent	Depth of price reduction
Inventory_Before_Markdown	Integer	Units exposed before action
Velocity_Before / After	Decimal	Weekly sales velocity response

KPI Definitions and Business Rules

Auditable merchandising measures

MERCHANDISING KPIS

KPI	DEFINITION
Net Sales	Units sold x selling price, after discount
Gross Margin \$	Net sales - cost of goods sold
Gross Margin %	Gross margin dollars / net sales
Sell-Through %	Units sold / (units sold + ending inventory)
Weeks of Supply	Ending inventory / average weekly unit sales
Stock Turn	Annualized cost of goods sold / inventory value
Sales Velocity	Units sold / active selling weeks
Average Unit Retail	Net sales / units sold
Markdown %	Discount value / original retail value
Sales per SKU	Category sales / active SKU count
Inventory Productivity	Gross margin dollars / average inventory investment
Plan Variance %	(Actual - plan) / plan

Synthetic Data Methodology

Generation, quality controls and intended use

GENERATION DESIGN

The dataset is deterministic and reproducible using seed 260812. Product demand follows category-specific velocity and inventory profiles, store demand multipliers, and seasonal factors. Commercial patterns were deliberately introduced to create realistic decisions: lighting productivity, premium decor exposure, seasonal allocation imbalance, vendor service variation and markdown response.

QUALITY CONTROLS

AREA	ROWS	CONTROL
Products	2,255	Unique SKU identifiers and complete product hierarchy
Stores	10	Unique locations across Atlantic and Central regions
Sales	120,000	Unique transaction identifiers and positive quantities
Inventory	22,550	Complete store-SKU snapshot coverage
Model status	PASS	Workbook allocation, OTB, transfer and score checks reconcile

INTENDED USE

Portfolio demonstration, SQL practice, Excel planning, dashboard prototyping and merchandise decision analysis. The data is fictional and must not be represented as actual retailer performance.